OWL Ontology for the College of Computing at Umm Al-Qura University

A project submitted for the Knowledge Representation and Inference course

COURSE INSTRUCTOR

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Content table

Task 1: Core class hierarchy (TBox)

Class	Description	Parent Class
CollegeOfComputing	Represents the College of Computing	owl:Thing
Activity	Academic or extracurricular activity	CollegeOfComputing
ResearchProject	Research project conducted in the college	Activity
Workshop	Educational workshop or event	Activity
Course	Academic course offered by a program	CollegeOfComputing
Department	Academic department within the college	CollegeOfComputing
Facility	Physical facility available in the college	CollegeOfComputing
ComputerLab	Computer laboratory	Facility
MeetingRoom	Meeting room	Facility
SmartClassroom	Technology-enabled classroom	Facility
Person	Any individual associated with the college	CollegeOfComputing
AdministrativeStaff	Administrative employee	Person
FacultyMember	Academic staffmember	Person
TeachingFaculty	Faculty member teaching a course	FacultyMember
ResearchActive	Faculty member supervising research	FacultyMember
Student	Student enrolled in a program	Person
ActiveStudent	Student enrolled in a program	Student
Program	Academic program offered by a department	CollegeOfComputing

Task 2: Object Properties

Property	Domain	Range	Relation Type	Characteristic
hasDepartment	CollegeOf Computing	Department	Organizational Relation	Inverse Of is Department Of
isDepartmentOf	Department	Collegeof Computing	Hierarchical Relation	Functional,Inverse
offersProgram	Department	Program	Organizational Relation	Object Property
teachesCourse	FacultyMember	Course	Participation Relation	Object Property
enrolledIn	Student	Program	Participation Relation	Functional
hasFacility	CollegeOfComputing	Facility	Support / Spatial Relation	Object Propety
participatesIn	Person	Activity	Participation Relation	Object Property
supervisesProject	FacultyMember	ResearchProject	Associative Relation	Object Property

Task 3: Data Properties

Property	**Domain**	**Range**	**Example**
hasName	owl:Thing	xsd:string	“Computer Lab 101”
hasID	Person	xsd:string	“S001”
hasEmail	Person	xsd:string	“nora@uqu.edu.sa”
hasCredits	Course	xsd:integer	3
hasCapacity	Facility	xsd:integer	40
hasDate	Activity	xsd:date	2026-05-20

Task 4: Axioms & Constraints

Type of Axiom	Class	Expression / Restriction	Explanation
Cardinality	Student	enrolledIn exactly 1 Program	A student must be enrolledIn in exactly one program
Disjointness	Student	Disjoint With FacultyMember	A person cannot be both a student and a faculty member
Disjointness	Student	Disjoint With AdministrativeStaff	A person cannot be both a student and administrative staff
Disjointness	FacultyMember	Disjoint With AdministrativeStaff	A person cannot be both faculty and administrative staff
Disjointness	Course	Disjoint With Program	A course cannot be a program
Disjointness	Department	Disjoint With Facility	A department cannot be a facility
Disjointness	Workshop	Disjoint With ResearchProject	A workshop cannot be a RsesearchProject
Disjointness	ComputerLab	Disjoint With MeetingRoom	A facility cannot be both a lab and a meeting room
Equivalence	ActiveStudent	Student and (enrolledIn some Program)	Any student enrolled in a program is automatically classified as an ActiveStudent
Equivalence	TeachingFaculty	FacultyMember and (teachesCourse some Course)	Any faculty member teaching a course is automatically classified as TeachingFaculty
Equivalence	ResearchActive	FacultyMember and (supervisesProject some ResearchProject)	Any faculty member supervising a research project is automatically classified as ResearchActive

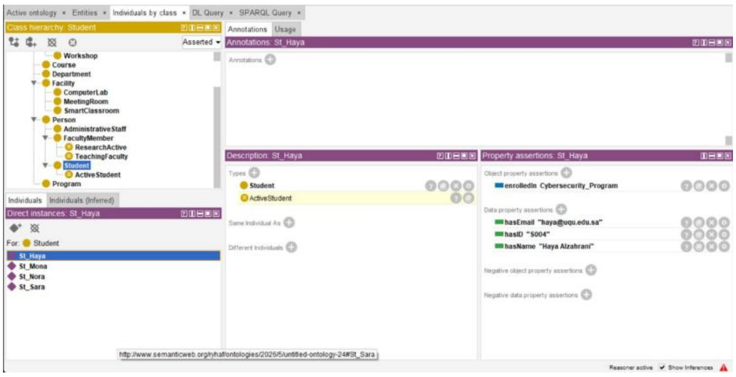
Note: Additional disjointness constraints were defined between academic and administrative roles to ensure ontology consistency. The reasoner automatically classifies individuals into ActiveStudent, TeachingFaculty, and ResearchActive based on the specified equivalent class definitions.

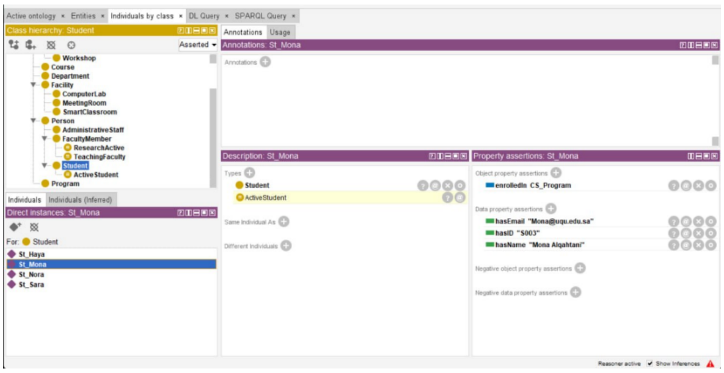
Task 5: Individuals (ABox)

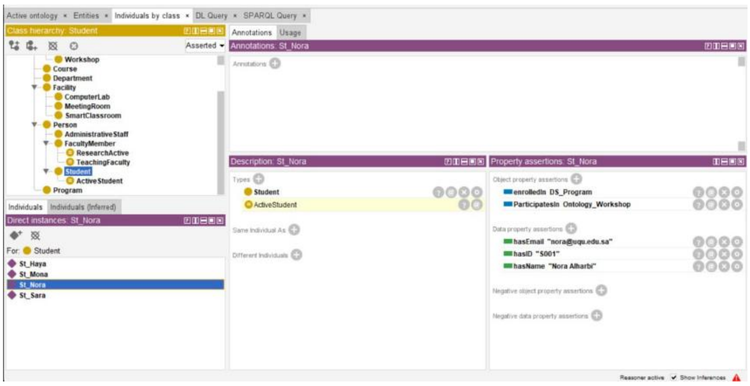
Individual	Class	Properties
UQU_CollegeOfComputing	CollegeOfComputing	hasDepartment, hasFacility
DS_Department	Department	offersProgram DS_Program
CS_Department	Department	offersProgram CS_Program
Cybersecurity_Department	Department	Offers program Cybersecurity_program
DS_Program	Program	
CS_Program	Program	
Cybersecurity_Program	Program	
KRI_Course	Course	hasCredits = 3
Database_Course	Course	hasCredits = 3
AI_Course	Course	hasCredits = 3
Dr_Huda	FacultyMember	teachesCourse KRI_Course, supervisesProject ontology_ ResearchProject
Dr_Maha	FacultyMember	teachesCourse Database_Course
Dr_Abeer	FacultyMember	teachesCourse AI_Course
St_Nora	Student	enrolledIn DS_Program, participatesIn ontologyWorkshop
St_Sara	Student	enrolledIn DS_Program, participatesIn ontologyWorkshop
St_Mona	Student	enrolledIn CS_Program
St_Haya	Student	enrolledIn Cybersecurity_Program
Lab_101	ComputerLab	hasCapacity = 35
MeetingRoom_201	MeetingRoom	hasCapacity = 20
SmartClassroom_301	SmartClassroom	hasCapacity = 40
Ontology_Workshop	Workshop	hasDate
Ontology_ResearchProject	ResearchProject	hasDate

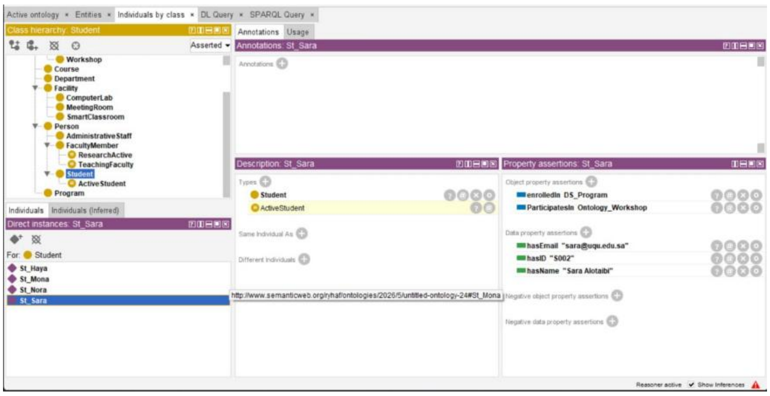
Task 6: Reasoning

ActiveStudent

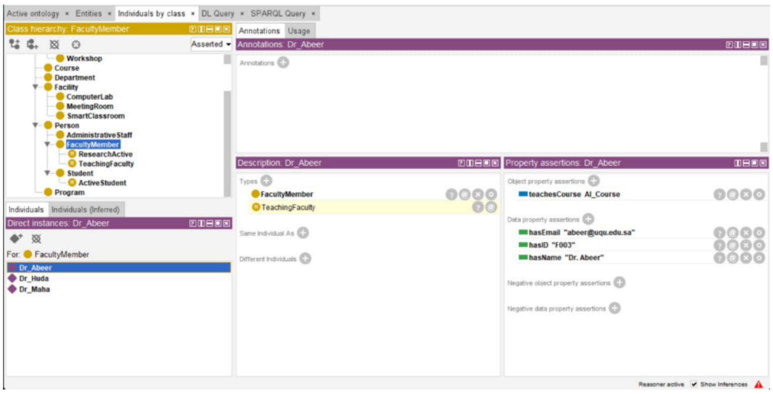


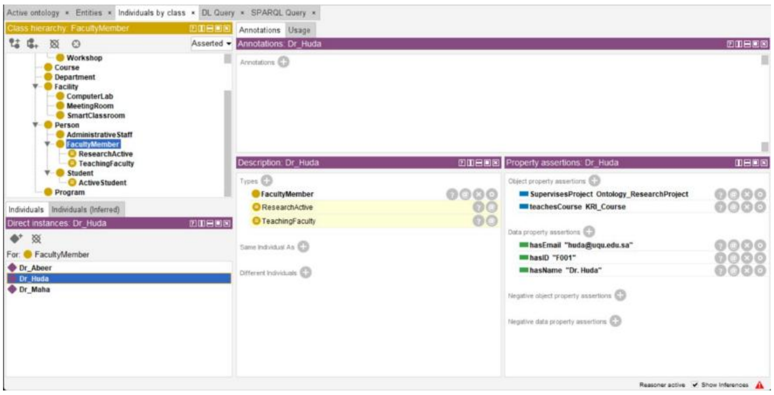


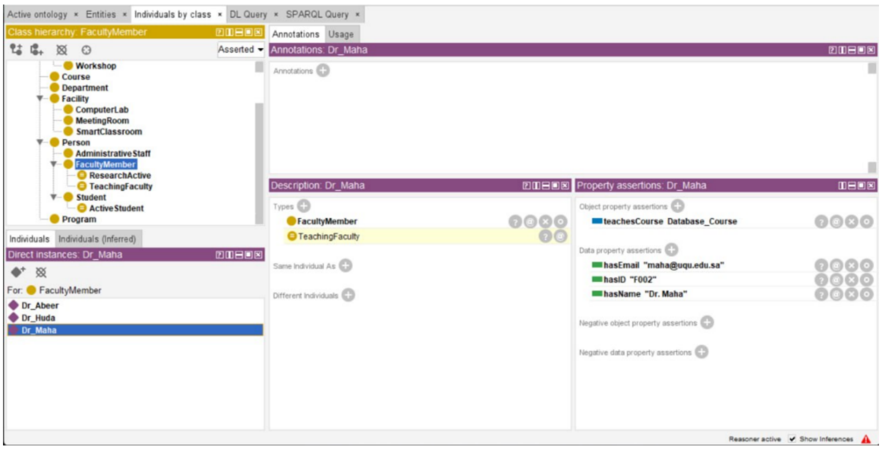




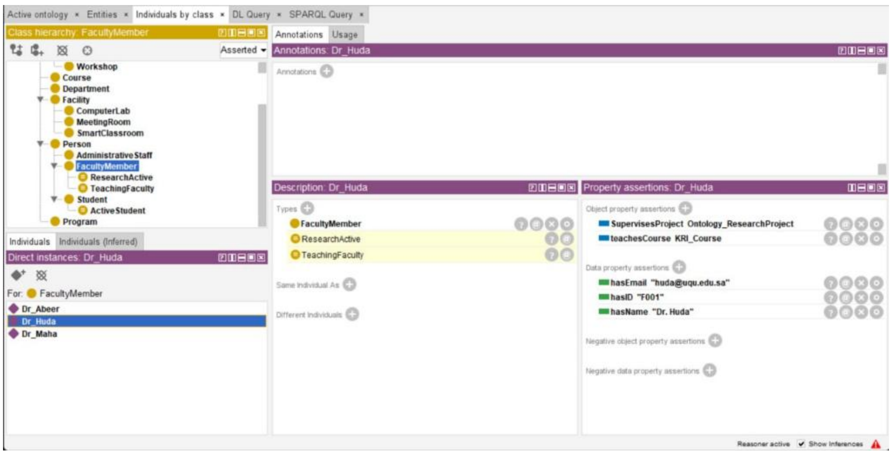
TeachingFacuity



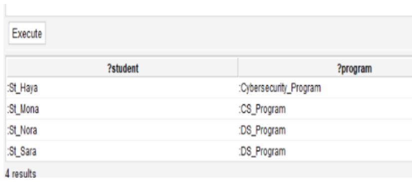
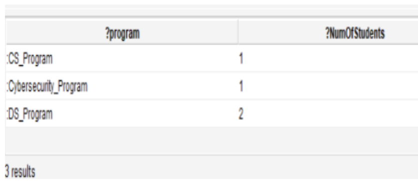
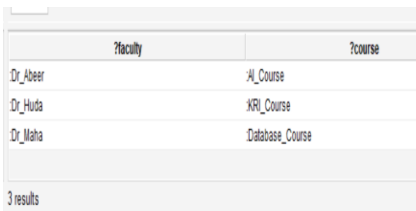




ResearchActive



Task 7: SPARQL Queries

Query #	Purpose	SPARQL Query	Output
1	Display students and their enrolled programs	SELECT ?student ?program WHERE { ?student :enrolledIn ?program .}	 Execute ?student ?program :St_Haya :Cybersecurity_Program :St_Mona :CS_Program :St_Nora :DS_Program :St_Sara :DS_Program 4 results
2	Count the number of students in each program using COUNT and GROUP BY.	SELECT ?program (COUNT(?student) AS ?NumOfStudents) WHERE { ?student :enrolledIn ?program .} GROUP BY ?program	 ?program ?NumOfStudents :CS_Program 1 :Cybersecurity_Program 1 :DS_Program 2 3 results
3	Display faculty members and their courses ordered by faculty name.	SELECT ?faculty ?course WHERE { ?faculty :teachesCourse ?course .} ORDER BY ?faculty	 ?faculty ?course :Dr_Abeer :AI_Course :Dr_Huda :KRI_Course :Dr_Maha :Database_Course 3 results
4	Display facilities with capacity greater than 25 using FILTER	SELECT ?facility ?capacity WHERE { ?facility :hasCapacity ?capacity. FILTER(?capacity > 25) }	 Execute ?facility ?capacity :Lab_101 35 :SmartClassroom_301 40 2 results